CS341 · COMPUTER GRAPHICS

Chapter 2 Vector Tools

Midterm summary

Muharib

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01 Point or vector?

Objects are represented mainly with linear primitives: points, lines and segments, planes and polygons. Two branches of mathematics are used: vector analysis and transformation.

Point	Vector
Specifies a location in space or in the plane	Has magnitude and direction, like velocity. It has no fixed location

Points specify **location** in space. Vectors have **magnitude and direction**. Points ≠ Vectors.

Coordinate systems: right-handed and left-handed (they differ in the direction of the z-axis).

02 The vector between two points

The **difference** between two points is a vector: **v = Q − P** (the vector from P to Q).
The **sum** of a point and a vector is a point: **P + v = Q**.

```
P = (1, 2),  Q = (4, 6)
```

```
v = Q − P = (4 − 1, 6 − 2) = (3, 4)
```

Asked before

Write the formula to compute a vector from point P to point Q. → **v = Q − P**

Mind the order: end minus start.

03 Operations on vectors

Two basic operations, both done componentwise:

```
Addition:     a = (3, 5, 8),  b = (−1, 2, −4)   →   a + b = (2, 7, 4)
```

```
Scalar:       a = (3, −5, 8), s = 5             →   5a = (15, −25, 40)
```

```
Subtraction:  a − c = a + (−c)
```

Linear combination

```
w = a1·v1 + a2·v2 + … + am·vm        (a1 … am are scalars)
```

Type	Condition	Example
Affine combination	`a1 + a2 + … + am = 1`	`3a + 2b − 4c` and `(1−t)a + t·b`
Convex combination	Sum = 1 and every `ai ≥ 0`	`0.3a + 0.2b + 0.5c`

Every convex combination is affine, but not the reverse (3 + 2 − 4 = 1, yet one coefficient is negative).

04 Length and the unit vector

```
|w| = √(w1² + w2² + … + wn²)
```

```
Unit vector:  â = a / |a|        (length = 1, used to refer to direction)
```

$a = (3, 4)$: $|a| = \sqrt{(9 + 16)} = 5$ $\quad \hat{a} = \left(\frac{3}{5}, \frac{4}{5}\right)$

- The process is called normalizing.
- A unit vector has length exactly 1.
- The standard unit vectors: $i = (1,0,0)$, $j = (0,1,0)$, $k = (0,0,1)$.

Asked in the midterm

A unit vector has a length greater than 1. → **False**

05 The dot product

$v \cdot w = v1 \cdot w1 + v2 \cdot w2 + \ldots + vn \cdot wn$ (the result is a number)

Property	
Symmetry	$a \cdot b = b \cdot a$
Linearity	$(a+c) \cdot b = a \cdot b + c \cdot b$
Homogeneity	$(sa) \cdot b = s(a \cdot b)$
Length	$\lvert b \rvert^2 = b \cdot b$

The angle between two vectors

$$\cos\theta = \hat{b} \cdot \hat{c} = \frac{b \cdot c}{|b|\ |c|}$$

```
b = (1, 0),  c = (1, 1)
```

```
b · c = 1          |b| = 1          |c| = √2
```

$$\cos\theta = \frac{1}{\sqrt{2}} \quad \rightarrow \quad \theta = 45°$$

The key rule

Two vectors are perpendicular (orthogonal, normal) if `b · c = 0`. A positive result means an acute angle; a negative one, an obtuse angle.

06 The 2D perp vector

Let a = (ax, ay). Then **a⊥ = (−ay, ax)** is the **counterclockwise perpendicular** to a.

```
a = (3, 4)   →   a⊥ = (−4, 3)
```

```
check: a · a⊥ = 3(−4) + 4(3) = 0  ✓
```

`−a⊥ = (ay, −ax)` is the clockwise perpendicular. Its applications include orthogonal projection and reflections.

07 The cross product

Also called the vector product. It is defined for 3D vectors only, and the result is a vector perpendicular to both.

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_x & a_y & a_z \\ b_x & b_y & b_z \end{vmatrix} = (a_y b_z - a_z b_y,\ a_z b_x - a_x b_z,\ a_x b_y - a_y b_x)$$

```
a = (1, 0, 0),  b = (0, 1, 0)
```

```
a × b = (0·0 − 0·1,  0·0 − 1·0,  1·1 − 0·0) = (0, 0, 1)
```

Application: finding the normal to a plane. Take three points `P1, P2, P3` on the plane, `a = P2 − P1` and `b = P3 − P1`; the normal is `a × b`.

08 Homogeneous representation

Relative to a coordinate frame with origin `O` and axes `a, b, c`, we add a fourth coordinate: 0 for a vector and 1 for a point.

```
vector v = (v1, v2, v3, 0)
```

```
point  P = (P1, P2, P3, 1)
```

Operation	Result
Point – Point	Vector
Point + Vector	Point
Vector + Vector	Vector
Scalar × Vector	Vector
Scalar × Vector + Scalar × Vector	Vector
Point + Point	**Not allowed (not a valid point)**

The reason in homogeneous form: the last coordinate of `P + R` is 1 + 1 = 2, which is neither 0 (vector) nor 1 (point).

09 Affine combinations of points

$$f \cdot P + g \cdot R = (f \cdot P1 + g \cdot R1,\ \ f \cdot P2 + g \cdot R2,\ \ f \cdot P3 + g \cdot R3,\ \ f + g)$$

$$\text{valid point} \ \Leftrightarrow \ f + g = 1$$

0.3P + 0.7R is a valid point, but P + R is not.

Any affine combination of points is a legitimate point.

$\frac{P1 + P2}{2}$ is a valid point: the midpoint.

Asked in the midterm

Sum two points results in a valid point. → **False**

A vector has both magnitude and direction. → **True**

The dot product of two perpendicular vectors is equal to zero. → **True**

Linear interpolation (tweening)

$$P = A + t \cdot v, \quad v = B - A$$

$$P = A + t(B - A) = t \cdot B + (1 - t) \cdot A$$

`t = 0` gives A, `t = 1` gives B and `t = 0.5` gives the midpoint.

$$A = (2, 2), \quad B = (6, 10), \quad t = 0.25$$

$$P = (2, 2) + 0.25(4, 8) = (3, 4)$$

10 Representing lines

Three kinds: a line (infinite both ways), a line segment (two endpoints) and a ray (a start and a direction).

Form	Data	Equation
Two-point form	{C, B}	Two points on the line
Parametric	{C, b}	`L(t) = C + b·t` where `b = B − C`
Point normal	{C, n}	`n · (R − C) = 0`

In the parametric form: `t = 0` at C, `t = 1` at B, `t > 1` beyond B and `t < 0` before C. Segment: `0 ≤ t ≤ 1`. Ray: `t ≥ 0`.

Converting between forms

```
Two-point → Parametric:      b = B − C
```

```
Parametric → Two-point:      B = C + b
```

```
Two-point → Point-normal:    n = (B − C)⊥
```

```
Point-normal → Two-point:    B = C − n⊥
```

```
Parametric ↔ Point-normal:   n = b⊥ ,   b = −n⊥
```

```
C = (1, 1),   B = (4, 3)
```

```
b = B − C = (3, 2)           L(t) = (1, 1) + t(3, 2)
```

```
n = b⊥ = (−2, 3)             −2(x − 1) + 3(y − 1) = 0
```

11 Planes in 3D

Form	Equation
Three-point form	Three non-collinear points
Parametric	`P(s, t) = C + a·s + b·t`
Point normal	`n · (R − B) = 0`

From three points, the normal comes from the cross product: `n = (P2 − P1) × (P3 − P1)`.

A planar patch is part of a plane: `P(s, t) = C + a·s + b·t` with `0 ≤ s, t ≤ 1`. The unit square in parameter space maps to a parallelogram in world coordinates.